

Nikolai Slavov

CONTACT INFORMATION

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Lab: slavovlab.net | Center: center.single-cell.net
ORCID ID: [0000-0003-2035-1820](https://orcid.org/0000-0003-2035-1820)

EDUCATION

Princeton University, Princeton, USA

Ph.D. in Molecular and Quantitative Cell Biology **06.2010**

M.A. in Molecular Biology **02.2006**

Mentor: David Botstein

Dissertation: Universality, specificity and regulation of *S. cerevisiae* growth rate response in different carbon sources and nutrient limitations

Massachusetts Institute of Technology (MIT), Cambridge, USA

B.S. in Biology **06.2004**

RESEARCH & ACADEMIC APPOINTMENTS

Parallel Squared Technology Institute (PTI), Boston, USA

Founding Director **2023–present**

Northeastern University, Boston, USA

Associate Professor, Department of Bioengineering **2021–present**

Faculty Fellow, Barnett Institute **2017–present**

Affiliated Faculty, Department of Chemistry and Chemical Biology **2016–present**

Affiliated Faculty, Department of Biology **2016–present**

Assistant Professor, Department of Bioengineering **2015–2021**

Harvard University, Cambridge, USA

Research Fellow, Departments of Systems Biology and Statistics **2013–2015**

Massachusetts Institute of Technology (MIT), Cambridge, USA

Research Fellow, van Oudenaarden laboratory, Departments of Biology and Physics **2011–2013**

AWARDS & FELLOWSHIPS

✓ [Discovery in Proteomic Sciences Award](#) **2022**

✓ College of Engineering Faculty Fellow **2022**

✓ [Allen Distinguished Investigator](#) (\$ 1, 500, 000) **2020**

✓ Excellence in Mentoring Award **2018**

✓ [NIH Director's New Innovator Award](#) (\$ 2, 352, 750) **2016**

✓ Broad Institute of Harvard and MIT SPARC Award **2014**

✓ Princeton University Dean's Award **2010**

✓ IRCSET Postgraduate Research Fellowship (72, 000 €) **2007**

✓ Finalist in the Young European Entrepreneur Competition **2006**

✓ Princeton Graduate Fellowship (\$ 45, 000) **2004**

- ✓ MIT Undergraduate Fellowship (\$ 125, 000) 2001
- ✓ Eureka Fellowship for Academic Excellence (10, 000 €) 2000
- ✓ Bronze Medal in the 31st International Chemistry Olympiad (*IChO*) 1999
- ✓ National Diploma for Exceptional Achievements in Chemistry 1999

SELECTED
PEER REVIEWED
PUBLICATIONS

[OA] – Open Access article ✉ – Corresponding author [Google Scholar Profile](#)

2023

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- ✓ **Slavov N**✉
Single-cell proteomics: quantifying post-transcriptional regulation during development with mass-spectrometry
Development (in press), 150, dev201492, DOI: [10.1242/dev.201492](https://doi.org/10.1242/dev.201492)
- ✓ van den Berg PR, Brenger-Currias NMLP, Budnik B, **Slavov N**✉, Semrau S
Integration of a multi-omics stem cell differentiation dataset using a dynamical model
PLOS Genetics **19** (5) :1-23 DOI: [10.1371/journal.pgen.1010744](https://doi.org/10.1371/journal.pgen.1010744) [OA]
- ✓ Huffman RG, Leduc A, Wichmann C, di Gioia M, Borriello F, Specht H, Derks J, Khan S, ..., **Slavov N**✉
Prioritized mass spectrometry increases the depth, sensitivity and data completeness of single-cell proteomics
Nature Methods, **20**, 714-722 DOI: [10.1038/s41592-023-01830-1](https://doi.org/10.1038/s41592-023-01830-1) [OA]
- ✓ Gatto L, Aebersold R, Cox J, Demichev V, Derks J, Emmott E, Franks AM, Ivanov AR, ..., **Slavov N**✉
Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments
Nature Methods **20**, 375386 DOI: [10.1038/s41592-023-01785-3](https://doi.org/10.1038/s41592-023-01785-3) [OA]
- ✓ MacCoss MJ, Alfaro J, Faivre, D, Wu C, Wanunu M, **Slavov N**✉
Sampling proteomes by emerging single-molecule and mass-spectrometry methods
Nature Methods **20**, 339346, DOI: [10.1038/s41592-023-01802-5](https://doi.org/10.1038/s41592-023-01802-5)
- ✓ **Slavov N**✉
Great Gains in Mass Spectrometry Data Interpretation
Journal of Proteome Research, **22** 3, 659 DOI: [10.1021/acs.jproteome.3c00099](https://doi.org/10.1021/acs.jproteome.3c00099)
- ✓ Derks J, **Slavov N**✉
Strategies for increasing the depth and throughput of protein analysis by plexDIA
Journal of Proteome Research **22** 697-705, DOI: [10.1021/acs.jproteome.2c00721](https://doi.org/10.1021/acs.jproteome.2c00721)

2022

2022

- ✓ Leduc A, Huffman RG, Cantlon J, Khan S, **Slavov N**[✉]
Exploring functional protein covariation across single cells using nPOP
Genome Biology, **23**, 261 DOI: [10.1186/s13059-022-02817-5](https://doi.org/10.1186/s13059-022-02817-5) [OA] ⇨ [Single-Cell Proteomics Bypasses Bottlenecks, Sways Skeptics](#), highlight by *GEN*
- ✓ Petelski AA, **Slavov N**, Specht H
Single-Cell Proteomics Preparation for Mass Spectrometry Analysis Using Freeze-Heat Lysis and an Isobaric Carrier
J. Vis. Exp., (190), DOI: [10.3791/63802](https://doi.org/10.3791/63802)
- ✓ Derks J, Leduc A, Wallmann G, Huffman G, Willetts M, Khan S, Specht H,..., **Slavov N**[✉]
Increasing the throughput of sensitive proteomics by plexDIA
Nature Biotechnology, DOI: [10.1038/s41587-022-01389-w](https://doi.org/10.1038/s41587-022-01389-w)
⇨ [Sensitive protein analysis with plexDIA](#), highlight by *Nature Methods*
⇨ [Uncovering Proteomic Patterns One Cell at a Time](#), highlight by *GEN*
- ✓ Specht H, **Slavov N**[✉]
Beyond Protein Sequence: Protein Isomerization in Alzheimer's Disease
Journal of Proteome Research 21 (2) :299-300, DOI: [10.1021/acs.jproteome.2c00016](https://doi.org/10.1021/acs.jproteome.2c00016)
- ✓ **Slavov N**[✉]
Counting protein molecules for single-cell proteomics
Cell 185 (2) :232-234, DOI: [10.1016/j.cell.2021.12.013](https://doi.org/10.1016/j.cell.2021.12.013)
- ✓ **Slavov N**[✉]
Learning from natural variation across the proteomes of single cells
PLoS Biology 20(1): e3001512, DOI: [10.1371/journal.pbio.3001512](https://doi.org/10.1371/journal.pbio.3001512) [OA]
- ✓ **Slavov N**[✉]
Scaling up single-cell proteomics
Molecular & Cellular Proteomics 21(1), 100179, DOI: [10.1016/j.mcpro.2021.100179](https://doi.org/10.1016/j.mcpro.2021.100179) [OA]

2021

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- ✓ Khoury L, **Slavov N**[✉]
Comprehensive Identification of Regulatory Protein Networks
Journal of Proteome Research 20(11): 4913-4914, DOI: [10.1021/acs.jproteome.1c00813](https://doi.org/10.1021/acs.jproteome.1c00813)
- ✓ He L, *et al.*, **Slavov N**, Marneros A,
Global characterization of macrophage polarization mechanisms and identification of M2-type polarization inhibitors
Cell Reports 37(5):109955, DOI: [10.1016/j.celrep.2021.109955](https://doi.org/10.1016/j.celrep.2021.109955)
- ✓ Petelski A, Emmott E, Leduc A, Huffman RG, Specht H, Perlman D, **Slavov N**[✉]
Multiplexed single-cell proteomics using SCoPE2
Nature Protocols 16, 5398–5425, DOI: [10.1038/s41596-021-00616-z](https://doi.org/10.1038/s41596-021-00616-z)
- ✓ **Slavov N**[✉]
Driving Single Cell Proteomics Forward with Innovation
Journal of Proteome Research 20, 11, 4915–4918, DOI: [10.1021/acs.jproteome.1c00639](https://doi.org/10.1021/acs.jproteome.1c00639)

- ✓ Slavov N[✉]
Measuring Protein Shapes in Living Cells
Journal of Proteome Research DOI: [10.1021/acs.jproteome.1c00376](https://doi.org/10.1021/acs.jproteome.1c00376)
- ✓ Slavov N[✉]
Increasing proteomics throughput
Nature Biotechnology, 39, 809–810, DOI: [10.1038/s41587-021-00881-z](https://doi.org/10.1038/s41587-021-00881-z)
- ✓ Slavov N[✉] *et al.*
Voices of biotech research: Single-cell proteomics
Nature Biotechnology, 39, 281286, DOI: [10.1038/s41587-021-00847-1](https://doi.org/10.1038/s41587-021-00847-1)
- ✓ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, Serra M, Kharchenko P, Koller A & Slavov N[✉]
Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity using SCoPE2
Genome Biology, 22, 50 DOI: [10.1186/s13059-021-02267-5](https://doi.org/10.1186/s13059-021-02267-5) [OA]
↪ [Towards resolving proteomes in single cells](#), highlight by *Nature Methods*

2020

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- ✓ Specht H, Slavov N[✉]
Optimizing accuracy and depth of protein quantification in experiments using isobaric carriers
Journal of Proteome Research, DOI: [10.1021/acs.jproteome.0c00675](https://doi.org/10.1021/acs.jproteome.0c00675)
- ✓ Petelski AA Slavov N[✉]
Analyzing ribosome remodeling in health and disease
Proteomics, 20, (17) :2000039 DOI: [10.1002/pmic.202000039](https://doi.org/10.1002/pmic.202000039)
- ✓ Slavov N[✉]
Single-cell protein analysis by mass-spectrometry
Current Opinion in Chemical Biology, 60, 1 - 9 DOI: [10.1016/j.cbpa.2020.04.018](https://doi.org/10.1016/j.cbpa.2020.04.018)
- ✓ Slavov N[✉]
Unpicking the proteome in single cells
Science, 367 (6477) :512 - 513, DOI: [10.1126/science.aaz6695](https://doi.org/10.1126/science.aaz6695)

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- ✓ Slavov N[✉] *et al.*
Voices in methods development: Single-cell proteomics
Nature Methods, 16, 945 - 951, DOI: [10.1038/s41592-019-0585-6](https://doi.org/10.1038/s41592-019-0585-6)
- ✓ Huffman RG, Chen AT, Specht H & Slavov N[✉]
DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
Journal of Proteome Research, 18, 6, 2493 - 2500, DOI: [10.1021/acs.jproteome.9b00039](https://doi.org/10.1021/acs.jproteome.9b00039)
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- ✓ Chen AT, Franks A & Slavov N[✉]
DART-ID increases single-cell proteome coverage
PLoS Computational Biology, 5(7):e1007082 DOI: [10.1371/journal.pcbi.1007082](https://doi.org/10.1371/journal.pcbi.1007082) [OA]
↪ [The single cell proteomics revolution](#), editorial highlight by *Bioanalysis Zone*

- ✓ Emmott EP, Jovanovic M & **Slavov N**[✉]
Approaches for studying ribosome specialization
Trends in Biochemical Sciences, 44(5):478 - 479, DOI: [10.1016/j.tibs.2019.01.008](https://doi.org/10.1016/j.tibs.2019.01.008)
- ✓ Emmott EP, Jovanovic M & **Slavov N**[✉]
Ribosome stoichiometry: from form to function
Trends in Biochemical Sciences, 44(2):95 - 109, DOI: [10.1016/j.tibs.2018.10.009](https://doi.org/10.1016/j.tibs.2018.10.009)
- ✓ Malioutov D., Chen T., Jaffe J., Airoidi E., Budnik B. & **Slavov N**[✉]
Quantifying homologous proteins and proteoforms
Molecular & Cellular Proteomics, 18(1):162 - 168, DOI: [10.1074/mcp.TIR118.000947](https://doi.org/10.1074/mcp.TIR118.000947)

2018

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- ✓ Specht H. & **Slavov N**[✉]
Transformative opportunities for single cell proteomics
Journal of Proteome Research, 17 (8), 2565 - 2571 , DOI: [10.1021/acs.jproteome.8b00257](https://doi.org/10.1021/acs.jproteome.8b00257)
↪ [Innovations in Proteomics: The Drive to Single Cells](#), editorial highlight
↪ [Through the Looking Glass of Single Cell Proteomics](#), highlight by *Technology Networks*
- ✓ Budnik B., Levy E., Harmange G. & **Slavov N**[✉]
SCoPE-MS: mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
Genome Biology, 19(1):161, DOI: [10.1186/s13059-018-1547-5](https://doi.org/10.1186/s13059-018-1547-5) [OA]
↪ [SCoPE-MS – We can finally do single cell proteomics!!!](#), highlight by *Proteomics News*
↪ [Researchers Apply Mass Spec to Single-Cell Proteomics](#), highlight by *GenomeWeb*
↪ [Single-cell proteomics](#) highlight by *the NIH Director's office*
↪ [Technology feature](#) highlight by *Nature Methods*
↪ [The single cell proteomics revolution](#), editorial highlight by *Bioanalysis Zone*
- ✓ Levy E. & **Slavov N**[✉]
Single cell protein analysis for systems biology
Essays in Biochemistry, 62(4):595 - 605, DOI: [10.1042/EBC20180014](https://doi.org/10.1042/EBC20180014)

2017

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- ✓ Franks A., Airoidi E.M., **Slavov N**[✉]
Post-transcriptional regulation across human tissues
PLoS Computational Biology, 13(5): e1005535, DOI: [10.1371/journal.pcbi.1005535](https://doi.org/10.1371/journal.pcbi.1005535) [OA]
↪ [How Statistics Weakened mRNAs Predictive Power](#) highlight by *The Scientist*
- ✓ Saleh D, Najjar M, Zelic M, Shah S, Nogusa S, Polykratis A, Paczosa MK, Gough PJ, Bertin J, Whalen M, Fitzgerald KA, **Slavov N**, Pasparakis M, Balachandran S, Kelliher M, Mecsas J, Degterev A.
Kinase Activities of RIPK1 and RIPK3 Can Direct IFN- β Synthesis Induced by Lipopolysaccharide.
The Journal of Immunology, 198(11):4435-4447, DOI: [10.4049/jimmunol.1601717](https://doi.org/10.4049/jimmunol.1601717)

2016

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- ✓ Klionsky, *et al.*, **Slavov N**, *et al.*
Guidelines for the use and interpretation of assays for monitoring autophagy
Autophagy, vol. 12, issue 1, 1–222, 12(1):1-222, DOI: [10.1080/15548627.2015.1100356](https://doi.org/10.1080/15548627.2015.1100356)

- ✓ Di Luca A, Hamill RM, Mullen AM, **Slavov N**, Elia G.
Comparative Proteomic Profiling of Divergent Phenotypes for Water Holding Capacity across the Post Mortem Ageing Period in Porcine Muscle Exudate
PLoS One, 11(3):e0150605, DOI: [10.1371/journal.pone.0150605](https://doi.org/10.1371/journal.pone.0150605) [OA]

2015

2015

- ✓ **Slavov N**✉
Making the most of peer review
eLife, 4:e12708, DOI: [10.7554/eLife.12708](https://doi.org/10.7554/eLife.12708) [OA]
- ✓ **Slavov N**✉, Semrau S., Airoidi E.M., Budnik B., van Oudenaarden A.
Differential stoichiometry among core ribosomal proteins
Cell Reports, vol. 13, issue 5, 865–873, DOI: [10.1016/j.celrep.2015.09.056](https://doi.org/10.1016/j.celrep.2015.09.056) [OA]
↔ [All Ribosomes Are Created Equal. Really?](#), highlight by *Trends in Biochemical Sciences*
↔ [New direction for tissue engineering and cancer therapeutics](#), highlight by *Science X*
- ✓ Alvarez JR, Zhiqiang B, Xu D, Yuan B, Lo KA, Yoon MJ, Lim YC, Knoll M, **Slavov N**, *et al.*
De-Novo Reconstruction of Adipose Tissue Transcriptomes Reveals Long Non-coding RNA Regulators of Brown Adipocyte Development
Cell Metabolism, vol. 21, issue 5, 764–776, DOI: [10.1016/j.cmet.2015.04.003](https://doi.org/10.1016/j.cmet.2015.04.003)

Started Slavov Lab at Northeastern University, August 2015

2014

2014

- ✓ Malioutov D.✉, **Slavov N**✉
Convex Total Least Squares
Journal of Machine Learning Research, W&CP, vol. 32, issue 1, 109–117 [OA]
- ✓ **Slavov N**✉, Budnik B., Schwab D., Airoidi E., van Oudenaarden A.✉
Constant Growth Rate Can Be Supported by Decreasing Energy Flux and Increasing Aerobic Glycolysis
Cell Reports vol. 7, issue 3, 705–714 [OA]

2013

2013

- ✓ **Slavov N**✉, Carey, J., Linse, S.✉
Calmodulin transduces Ca^{+2} oscillations into differential regulation of its target proteins
ACS Chemical Neuroscience, vol. 4, 601–612 [OA]
- ✓ **Slavov N**✉, Botstein D.✉
Decoupling Nutrient Signaling from Growth Rate Causes Aerobic Glycolysis and Dereglulation of Cell Size and Gene Expression
Molecular Biology of the Cell, vol. 24, issue 2, 157–168 [OA]

2012

2012

- ✓ **Slavov N**[✉], van Oudenaarden A.[✉]
How to Regulate a Gene: To Repress or to Activate?
Molecular Cell, vol. 46, issue 5, 551–552 [\[OA\]](#)
- ✓ **Slavov N**[✉], Airoidi E.M., van Oudenaarden A., Botstein D.[✉]
A Conserved Cell Growth Cycle Can Account for the Environmental Stress Responses of Divergent Eukaryotes
Molecular Biology of the Cell, vol. 23, no. 10, 1986–1997 [\[OA\]](#)

2011

2011

- ✓ **Slavov N**[✉], Macinskas J., Caudy A., Botstein D.[✉]
Metabolic Cycling without Cell Division Cycling in Respiring Yeast
PNAS, vol. 108, no. 47, 19090–19095 [\[OA\]](#)
- ✓ **Slavov N**[✉], Botstein D.[✉]
Coupling among Growth Rate Response, Metabolic Cycle and Cell Division Cycle in Yeast
Molecular Biology of the Cell, vol. 22, no. 12, 1997–2009 [\[OA\]](#)

2010

2010

- ✓ **Slavov N**[✉]
Inference of Sparse networks with Unobserved Variables. Application to Gene Regulatory Networks
Journal of Machine Learning Research, W&CP, vol. 9, 757–764 [\[OA\]](#)
- ✓ **Slavov N**[✉], Botstein, D.
Universality, Specificity and Regulation of *S. cerevisiae* Growth Rate Response in Different Carbon Sources and Nutrient Limitations
Dissertation, Princeton University [\[OA\]](#)
- ✓ Silverman SJ., **Slavov N**, Petti A., Parsons L., Briehof R., Thiberge S., Zenklusen D., Gandhi SJ., Larson D., Singer R., Botstein D.[✉]
Metabolic cycling in single yeast cells from unsynchronized steady-state populations limited on glucose or phosphate
PNAS, vol. 107, no. 15, 6946–6951 [\[OA\]](#)

2009 and before

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- ✓ **Slavov N**[✉], Dawson KA. (2009)
Correlation Signature of the Macroscopic States of the Gene Regulatory Network in Cancer
PNAS, vol. 106, no. 11, 4079–4084 (“In This Issue” article) [\[OA\]](#)
- ✓ Tagkopoulos I.[✉], **Slavov N**[✉], Kung S.[✉] (2005)
Multi-Class Biclustering and Classification Based on Modeling of Gene Regulatory Networks
5th IEEE Symposium on Bioinformatics and Bioengineering (BIBE’05).

PATENTS

- **Slavov N[✉]**, Budnik B., Specht H., Levy E. Mass spectrometry technique for single cell proteomics, 16/251,039 (2019)

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PREPRINTS

[GOOGLE SCHOLAR](#)

- ▷ Wallmann G., Leduc A., **Slavov N[✉]** (2023)
Data-Driven Optimization of DIA Mass-Spectrometry by DO-MS
bioRxiv, DOI: [10.1101/2023.02.02.526809](https://doi.org/10.1101/2023.02.02.526809) [\[OA\]](#)
- ▷ Derks J, **Slavov N[✉]** (2022)
Strategies for increasing the depth and throughput of protein analysis by plexDIA
bioRxiv, DOI: [10.1101/2022.11.05.515287](https://doi.org/10.1101/2022.11.05.515287) [\[OA\]](#)
- ▷ Gatto L, ..., **Slavov N[✉]** (2022)
Initial recommendations for performing, benchmarking, and reporting single-cell proteomics experiments
arXiv, DOI: [10.48550/arXiv.2207.10815](https://doi.org/10.48550/arXiv.2207.10815) [\[OA\]](#)
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Sampling proteomes by emerging single-molecule and mass-spectrometry methods
arXiv, DOI: [10.48550/arXiv.2208.00530](https://doi.org/10.48550/arXiv.2208.00530) [\[OA\]](#)
- ▷ Huffman RG, Leduc A, ..., **Slavov N[✉]** (2022)
Prioritized single-cell proteomics reveals molecular and functional polarization across primary macrophages
bioRxiv, DOI: [10.1101/2022.03.16.484655](https://doi.org/10.1101/2022.03.16.484655) [\[OA\]](#)
- ▷ Derks J, Leduc A, Huffman RG, Specht H, Ralser M, Demichev V, **Slavov N[✉]** (2021)
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bioRxiv, DOI: [10.1101/2021.11.03.467007](https://doi.org/10.1101/2021.11.03.467007) [\[OA\]](#)
- ▷ Leduc A, Huffman RG, **Slavov N[✉]** (2021)
Droplet sample preparation for single-cell proteomics applied to the cell cycle
bioRxiv, DOI: [10.1101/2021.04.24.441211](https://doi.org/10.1101/2021.04.24.441211) [\[OA\]](#)
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- ▷ Petelski AA, **Slavov N** (2020)
Analyzing ribosome remodeling in health and disease
arXiv, [arXiv:2007.05839](https://arxiv.org/abs/2007.05839) [OA]
- ▷ **Slavov N** (2020)
Single-cell protein analysis by mass-spectrometry
arXiv, [arXiv:2004.02069](https://arxiv.org/abs/2004.02069) [OA]
- ▷ Specht H, Emmott E, Petelski A, Huffman RG, Perlman D, *et al.* & **Slavov N** (2019)
Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity *bioRxiv*, DOI: [10.1101/665307](https://doi.org/10.1101/665307) [OA]
↪ [Over 1,000 SINGLE CELL PROTEOMES!](#), highlight by *Proteomics News*
↪ [Technology feature](#) highlight by *Nature Methods*
- ▷ Specht H, Emmott E, Koller A, **Slavov N** (2019)
High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
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- ▷ Huffman G, Specht H, Chen AT, **Slavov N** (2019)
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- ▷ Specht H, Harmange G, Perlman DH, Emmott E, Niziolek Z, Budnik B, **Slavov N** (2018)
Automated sample preparation for high-throughput single-cell proteomics
bioRxiv, DOI: [10.1101/399774](https://doi.org/10.1101/399774) [OA]
- ▷ Chen A, Franks A, **Slavov N** (2018)
DART-ID increases single-cell proteome coverage
bioRxiv, DOI: [10.1101/399121](https://doi.org/10.1101/399121) [OA]
- ▷ Levy E. & **Slavov N** (2018)
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- ▷ Specht H. & **Slavov N** (2018)
Routinely quantifying single cell proteomes: A new age in quantitative biology and medicine
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- ▷ Berg P., Budnik B., **Slavov N**[✉], Semrau S. (2017)
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bioRxiv, DOI: [10.1101/102681](https://doi.org/10.1101/102681) [OA]
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↔ [Researchers Apply Mass Spec to Single-Cell Proteomics](#), highlight by *GenomeWeb*
↔ [Interview](#), highlight by *Front Line Genomics*
- ▷ **Slavov N**[✉] (2015)
From differential transcription of ribosomal proteins to differential structure of ribosomes
PeerJ Preprint, DOI: [10.7287/peerj.preprints.1504v1](https://doi.org/10.7287/peerj.preprints.1504v1) [OA]
- ▷ Franks A., Airoidi E.M., **Slavov N**[✉] (2015)
Post-transcriptional regulation across human tissues
bioRxiv, DOI: [10.1101/020206](https://doi.org/10.1101/020206) [OA]
- ▷ **Slavov N**[✉], Botstein, D., Caudy A. (2014)
Extensive regulation of metabolism and growth during the cell division cycle
bioRxiv, DOI: [10.1101/005629](https://doi.org/10.1101/005629) [OA]
- ▷ **Slavov N**[✉], Semrau S., Airoidi E.M., Budnik B., van Oudenaarden A. (2014)
Differential stoichiometry among core ribosomal proteins
bioRxiv, DOI: [10.1101/005553](https://doi.org/10.1101/005553) [OA]

OPINIONS &
POPULAR SCIENCE
PUBLICATIONS

- ▷ **Slavov N**[✉] (2018) We Must Demand Evidence of Peer Review
The Scientist, [Link](#)
- ▷ **Slavov N**[✉] (2017) What was rate limiting was the idea, not the technology
Front Line Genomics, [Link](#)
- ▷ **Slavov N**[✉] (2015) Making the most of peer review
eLife, [Link](#)
- ▷ **Slavov N**[✉] (2014) Accomplishments Over Accolades!
The Scientist, [Link](#)
- ▷ **Slavov N**[✉] (2014) Discoveries lie hidden behind the façade of popular assumptions
Cell Reporter, [Link](#)

- ▷ Slavov N[✉] (2013) The Best Projects Are Least Obvious
The Scientist, [Link](#)
- ▷ Slavov N[✉] (2013) A Wonderful Christmas with Mysterious Oscillations
PubChase, [Link](#)
- ▷ Slavov N[✉] (2012) The Mission of MIT
The MIT Tech, [Link](#)

INVITED KEYNOTE
TALKS

- | | |
|--|---------------------|
| 1. Driving discoveries with single-cell proteomics
<i>Summer School on Computational Proteomics</i> | June.27.2023 |
| 2. Strategies for increasing the depth and throughput of single-cell proteomics
<i>ASMS Congress</i> | June.4.2023 |
| 3. Single-cell proteomics: An exhilarating journey
<i>HUPO 2022 World Congress, Human Proteome Organization</i> | Dec.7.2022 |
| 4. Single-Cell Protein Analysis by Mass-Spectrometry, Video
<i>ASMS Asilomar Conference on Single-Cell Mass Spectrometry</i> | Oct.8.2022 |
| 5. Scaling up single-cell analysis to the proteome
<i>IMSC 2022 Conference</i> | Aug.29.2022 |
| 6. Exploring functional protein covariation across single cells
<i>Next-Generation Protein Analysis and Detection</i> | May.30.2022 |
| 7. Learning from natural variation across the proteomes of single cells
<i>XIV Annual Congress of the European Proteomic Association</i> | Apr.4.2022 |
| 8. Learning from natural variation across the proteomes of single cells
<i>3rd IBSE International Symposium</i> | Feb.3.2022 |
| 9. Increasing the sensitivity, reliability, reproducibility and throughput of single-cell proteomics
<i>HUPO Reconnect 2021 World Congress, Human Proteome Organization</i> | Nov.15.2021 |
| 10. Nano proteomic sample preparation for single-cell protein analysis
<i>International Symposium on Microscale Separations and Bioanalysis</i> | Jul.12.2021 |
| 11. Droplet sample preparation for single-cell proteomics applied to the cell cycle
<i>2021 AOHUPO, Human Proteome Organization</i> | Jun.30.2021 |
| 12. Computational opportunities for single-cell proteomics, Video
<i>MaxQuant Summer School</i> | Jun.25.2021 |
| 13. Mapping the transcriptome and proteome of tissues in 3D at single-cell resolution
<i>Plant Cell Atlas</i> | Apr.08.2021 |
| 14. Characterizing transcriptional and post-transcriptional regulation from single-cell variability
<i>CELLINK Partnership Conference</i> | Dec.11.2020 |
| 15. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity (link)
<i>The 2020 HUPO World Congress, Human Proteome Organization</i> | Oct.20.2020 |
| 16. Single-cell proteomics by mass-spectrometry (link)
<i>Netherlands Proteomics Platform, Utrecht, Netherlands</i> | Jan.17.2020 |

17. Quantifying proteins in single cells at high-throughput
Single-cell Omics Symposium at Peking University **Oct.20.2019**
18. High-throughput single-cell proteomics quantifies the emergence of macrophage heterogeneity
European Single Cell Proteomics Conference, Institute of Molecular Pathology (IMP) **Aug.27.2019**
19. Quantifying proteins in single cells at high-throughput
The 28th International KOGO Annual Conference, South Korea **Sept.06.2019**
20. DO-MS: Data-Driven Optimization of Mass Spectrometry Methods
MaxQuant Summer School, University of Wisconsin– Madison, [Video](#) **Jul.25.2019**

INVITED
DEPARTMENTAL
TALKS & VISITS

1. Biological and technological frontiers for single-cell proteomics
MIT **Dec.05.2023**
2. Biological and technological frontiers for single-cell proteomics
Yale University **Sept.26.2023**
3. Biological and technological frontiers for single-cell proteomics
University of Nebraska Medical Center **June.16.2023**
4. Exploring functional protein covariation across single cells
Department of Biochemistry and Molecular Biology, UC Riverside **April.25.2023**
5. Exploring functional protein covariation across single cells
Institute for Systems Biology **April.21.2023**
6. Exploring functional protein covariation across single cells
Department of Pharmacology at UC San Diego **Feb.27.2023**
7. Driving biology with single-cell proteomics
Eli Lilly **Nov.11.2022**
8. Functional protein covariation across single cells
Systems Biology Division, NIGMS, NIH **Nov.10.2022**
9. Driving biology with single-cell proteomics: New data acquisition and interpretation methodologies,
[Video](#)
2022 IMaSS Single Cell Day **Oct.12.2022**
10. Investigating biological mechanisms with high-throughput single-cell proteomics
Takeda **Sep.13.2022**
11. Exploring functional protein covariation across single cells, [Video](#)
GIGA University of Liege, Belgium **Sept.2.2022**
12. Exploring functional protein covariation across single cells
Laboratory of Receptor Biology and Gene Expression, NCI, NIH **Mar.9.2022**
13. Single-cell proteomic and transcriptomic analysis of macrophage polarization
NYU Pathology Department **Feb.14.2022**
14. Understanding biological systems: In search of direct causal mechanisms
D. E. Shaws **Jan.19.2022**
15. Single-cell proteomic and transcriptomic analysis of macrophage polarization
Oxford University and Charité' **Oct.20.2021**

16. Using Single-Cell Proteomics and Transcriptomics to Understand Gene Regulation
University of Texas Medical Branch **Sept.16.2021**
17. Single-cell proteomic and transcriptomic analysis of macrophage polarization
Department of Cell Biology, Duke University **Jan.14.2021**
18. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity
Peking University **Oct.11.2020**
19. Single-cell proteomic and transcriptomic analysis of macrophage heterogeneity
Department of Biochemistry, Boston University **Sept.14.2020**
20. Quantifying proteins in single cells at high-throughput
Center for Proteomics and Metabolomics, Leiden University **Jan.16.2020**
21. Quantifying proteins in single cells at high-throughput
Biotech Research and Innovation Center, University of Copenhagen **Jan.15.2020**
22. Coordination among metabolism and cell division
Boston Children Hospital & Harvard Medical School **Nov.14.2019**
23. Single-cell proteomics quantifies the emergence of macrophage heterogeneity
Department of Physiology, Johns Hopkins University School of Medicine **Nov.6.2019**
24. Understanding biological systems: In search of direct causal mechanisms ([link](#)) [Video](#)
Models, Inference & Algorithms Initiative (MIA), Broad Institute of MIT & Harvard **Apr.3.2019**
25. Understanding and controlling cell differentiation using single-cell mass-spectrometry ([link](#))
Department of Systems Biology (Theory Lunch), Harvard Medical School **May.18.2018**
26. Principles of ribosome-mediated translational regulation
The Center for Engineering in Medicine, Massachusetts General Hospital **Mar.17.2017**
27. Principles of ribosome-mediated translational regulation
Systems Biology Seminar, Boston University **Oct.13.2016**
28. Ribosome-mediated translational regulation: Direct methods for quantifying the ribosome code
Center for Systems Biology, Columbia University **Apr.20.2016**
29. Coordinating metabolism and ribosome-mediated translational regulation
Department of Biology, New York University **Apr.19.2016**
30. Quantifying protein isoforms ([link](#))
Models, Inference & Algorithms Initiative (MIA), Broad Institute **Sept.14.2015**

INVITED
PRESENTATIONS
FOR FUNDING

RECOMMENDATIONS

- Opportunities for single-cell proteomics and metabolomics
CZI **Oct.6.2022**
- Driving single-cell proteomics forward with innovation
National Institute on Aging, NIH **Sept.16.2021**
- Current state of single-cell proteomics, standards, potentials and quantitative benchmarks
Functional Proteomics, HubMap & Common Fund, NIH **Oct.8–9.2020**

- Single-cell proteomics
Helmsley charitable trust, Gut cell atlas meeting, NYC **Oct.28.2019**
- Transformative opportunities for single-cell proteomics
NHLBI Single-cell workshop, National Institutes of Health (NIH) **Oct.17.2019**

INVITED
CONFERENCE TALKS

1. Single Cell Proteomics. Parallel Squared.
Allen Frontiers Group **April.21.2023**
2. High-throughput Single Cell Proteomics
SLAS2023 **April.21.2023**
3. Exploring functional protein covariation across single cells
The Scientist: Pushing Towards Single-Cell Resolution **Dec.12.2022**
4. Introduction to single-cell proteomics by mass-spectrometry, [Recorded video](#)
HUPO 2022 World Congress, Human Proteome Organization **Dec.4.2022**
5. Prioritized single-cell proteomics reveals molecular and functional polarization across primary macrophages
The 51st Annual Meeting of the Japanese Society for Immunology **Dec.6.2022**
6. Global analysis of post-transcriptional regulation with single-cell resolution
Tri-Omics, Oxford Global **Sept.28.2022**
7. Driving biology with single-cell proteomics: New data acquisition and interpretation methodology for sensitive protein analysis
Italian Mass Spectrometry Society (IMaSS) **Oct.12.2022**
8. Innovations in single-cell proteomics enable exploring functional protein covariation across thousands of single cells
International Symposium on MS in the Health and Life Sciences **Aug.14.2022**
9. Exploring functional protein covariation across single cells using parallel proteomics
Gordon Research Conference: Single-Cell Genomics **May.2.2022**
10. Single-cell proteomics
The Conference on Retroviruses and Opportunistic Infections **Feb.13.2022**
11. Droplet sample preparation for single-cell proteomics applied to the cell cycle
CZI Seed Network Webinar **Jul.20.2021**
12. Single-cell Proteomic and Transcriptomic Analysis of Macrophage Heterogeneity ([link](#))
HUPO World Congress **Oct.20.2021**
13. Single-cell protein analysis by mass-spectrometry ([link](#))
The future of single cell analysis, Earlham Institute **Feb.16.2020**
14. Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity ([link](#))
Applied Bioinformatics in Life Sciences **Feb.14.2020**
15. Single-cell mass-spectrometry quantifies the emergence of macrophage heterogeneity ([link](#))
Cellular and Molecular Bioengineering Conference, BMES **Jan.3.2020**
16. Single-cell proteomics for studying regeneration and aging ([link](#))
MDI Biological Symposium for Regeneration and Aging **Oct.9.2019**
17. Mapping the Transcriptome and Proteome of Human Testis in 3D
CZI Human Cell Atlas Seed Networks Meeting **Jul.30.2019**

18. Transformative opportunities for single cell proteomics
The Greater Boston Mass Spectrometry Discussion Group **Dec.13.2018**
19. Understanding and controlling cell differentiation based on comprehensive mass-spec quantification of proteins and signaling in single cells
UPENN Single Cell Biology Symposium, [Video](#) **Nov.13.2018**
20. Chair for the Single Cell Proteomics session of HUPO
Automated sample preparation for high-throughput single-cell proteomics **Oct.3.2018**
21. Using single-cell proteomics to engineer directed cell differentiation
Bioengineering & Translational Medicine Conference **Sept.27.2018**
22. Analyzing single cell proteomics data
ETH, Biognosys Discovery Proteomics Seminar **Jul.7.2018**
23. Mass-spectrometry of single mammalian cells quantifies proteome heterogeneity during cell differentiation
CSHL Meeting: Single Cell Analyses **Nov.10.2017**
24. Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Annual Single Cell Analysis USA Congress, Boston **Oct.23.2017**
25. Quantifying protein heterogeneity during the differentiation of single stem cells
Global CESI-MS Symposium **Oct.06.2017**
26. Single cell proteomics quantifies proteome heterogeneity during cell differentiation
Festival of Genomics, Boston **Oct.04.2017**
27. From differential transcription of ribosomal proteins to differential structure of ribosomes
Genetics, Genomics and Beyond, Calico **Sept.16.2017**
28. Differential stoichiometry among core ribosomal proteins
CSHL Translational Control, Cold Spring Harbor Laboratory **Sept.08.2016**
29. Principles of ribosome-mediated translational regulation
Center for Interdisciplinary Research on Complex Systems **Apr.12.2016**
30. Coordination among cell growth, cell division and gene expression
Broad Institute of MIT and Harvard, YSB **Dec.9.2015**
31. Coordinating cell growth and metabolism with ribosome-mediated translational regulation
Northeastern University, Biology Department Colloquium **Oct.19.2015**
32. Direct quantification of highly homologous proteins by mass-spectrometry indicates physiological changes in the protein composition of the ribosome
Broad Institute of MIT and Harvard, CBBO **Jun.25.2014**
33. Variable stoichiometry among core ribosomal proteins
Broad Institute of MIT and Harvard **Jun.19.2014**
34. Quantification of translational regulation of homologous proteins with unprecedented accuracy
Harvard HMS Systems Biology Theory Lunch **Mar.21.2014**
35. Exponentially growing cells can support a constant growth rate by very different metabolic strategies
Fourth Annual Physical Sciences in Oncology Meeting, NCI **Apr.18.2013**
36. Experimentally identified trade-offs of aerobic glycolysis
Center of Cancer Systems Biology at Tufts **Dec.18.2012**
37. Trade-offs of aerobic glycolysis
Harvard HMS Systems Biology Theory Lunch **Sep.12.2012**

38. The dynamics of exponential cell growth at constant growth rate
MIT Physical Sciences-Oncology Center **Jun.11.2012**
39. Metabolic cycling without cell division cycling
Bauer Forum at Harvard FAS Systems Biology **Apr.15.2011**

CONFERENCE TALKS

1. Quantifying homologous proteins and proteoforms
US HUPO Conference **Mar.14.2016**
2. Variable stoichiometry among core ribosomal proteins
2014 ASCB/IFCB Annual Meeting **Dec.8.2014**
3. The dynamics of exponential growth
GSA Yeast Meeting, Princeton University **Aug.5.2012**
4. The dynamics of exponential growth
Broad Institute of Harvard and MIT **Apr.17.2012**
5. Inference of Sparse Networks with Unobserved Variables. Application to Gene Regulatory Networks
International Conference on Artificial Intelligence and Statistics **May.14.2010**

COMPLETED FUNDING

- ✓ NIH Director's New Innovator Award (\$ 2, 352, 750) **2016-2021**
Sole PI, 100 % DP2GM123497, Ribosome-mediated translational regulation during stem cell differentiation
- ✓ Merck Exploratory Science Center Fellowship (\$ 242, 827) **2019-2020**
PI, 100 %, Developing next generation single-cell proteomics and using it to characterize the functional polarization of macrophages
- ✓ Sanofi iAward (\$ 196, 000) **2018**
PI, 100 %, Developing a technology platform for discovering biomarkers and drug targets
- ✓ NEU Tier I Award (\$ 50, 000) **2016**
PI, 50 %, Internal funding
- ✓ Broad Institute of Harvard and MIT SPARC Award (\$ 100, 000) **2014**
PI, 100 %
- ✓ IRCSET Postgraduate Research Fellowship (72, 000 €) **2007**
PI, 100 %

TEACHING & ADVISING

Courses Taught

Teaching websites

- Principles of Bioengineering**, Northeastern University **2020 – 2020**
 Helping introduce doctoral students to research, [Website](#)
- Method and Logic in Systems Biology and Bioengineering**, Northeastern University **2019 – 2021**
 Discussion of seminal papers, [Website](#)
- Mathematical Methods for Engineers**, Northeastern University **2016 – 2021**
 Introduction to major concepts and methods in linear algebra differential equations, [Website](#)

Method and Logic in Quantitative Biology , Princeton University	2009 – 2010
Introduction to major ideas and concepts in quantitative biology based on primary literature	
Residential Graduate Student at Forbes College , Princeton University	2009 – 2010
Advised students; organized language tables, sport events and trips	
Integrated Quantitative Introduction to Natural Sciences , Princeton University	2006 – 2010
Interdisciplinary curriculum including physics, computer science, biochemistry, genetics, physiology and emphasizing quantitative problem solving and the connections among these disciplines	
Terrascope , MIT	2002 – 2003
Selected for two subsequent years to lead undergraduate student groups working on complex real-world problems	
Biochemistry , MIT	2003 – 2004
Taught precepts and helped undergraduate students prepare for examinations	

Research associates at Northeastern University	Alumni website
Dr. Edward Emmott	<i>Currently Tenure-track professor at Liverpool University</i>
Dr. Toni Koller	<i>Currently Director of Koch Proteomics Facility at MIT</i>
Dr. David Perlman	<i>Currently Senior Principal Scientist at Merck</i>
Dr. Harrison Specht	<i>Currently CSO of ParallelSq</i>
Dr. Sina Rafiee,	<i>2022 – current</i>
Dr. Jason Derks,	<i>2023 – current</i>

Doctoral Students at Northeastern University	Group members website
Harrison Specht, Bioengineering PhD student	<i>2016 – 2022</i>
Aleksandra Petelski, Bioengineering PhD student	<i>2017 – 2022</i>
R. Gray Huffman, Bioengineering PhD student	<i>2017 – 2022</i>
Jason Derks, Bioengineering PhD student	<i>2019 – 2022</i>
Tianchi Chen, Bioengineering PhD student (Transferred to E. Sontag in 2018)	<i>2016 – 2022</i>
Andrew Leduc, Bioengineering PhD student	<i>2019; expected graduation: 2024</i>
Shiri Tsour, Bioengineering PhD student	<i>2019; expected graduation: 2024</i>
Luke Khoury, Bioengineering PhD student	<i>2021; expected graduation: 2025</i>
Saad Khan, Bioengineering PhD student	<i>2021; expected graduation: 2025</i>
Burak Yilmaz, Bioengineering PhD student	<i>2021; expected graduation: 2026</i>
Hannah Harens, Bioengineering PhD student	<i>2022; expected graduation: 2027</i>
Orhun Kok, Bioengineering PhD student	<i>2022; expected graduation: 2027</i>
Satyajit Rajapurkar, Bioengineering PhD student	<i>2022; expected graduation: 2027</i>

Masters Students at Northeastern University	Group members website
Joseph Collura, Bioengineering MS student	<i>2020 – 2021</i>
Collin Gwilt, Bioengineering MS student	<i>2020 – 2022</i>
Georg Wallmann, MS student at LMU Munich	<i>2021 – 2022</i>
Widmer Simon, MS student at ETH	<i>2022; expected graduation: 2023</i>

Undergraduate students at Northeastern University

[Alumni website](#)

Ezra Levy

Next position: PhD student at Princeton University

Guillaume Harmange

Next position: PhD student at University of Pennsylvania

Erin Provost

Next position: Research Associate I at Atalanta Therapeutics

Albert Chen

Next position: Research Associate the Broad Institute

Kaely Gallagher

Next position: Engineer at Sana Biotechnology, Inc.

Ryan Posey

Next position: PhD student at Harvard University

Aanie Phillips

Next position: Medical student at Baylor College of Medicine

Matthew Dowling

Next position: PhD student at Rochester University

Levon Rodriguez

Currently Northeastern undergraduate student

Sebastian Hymson

Currently Northeastern undergraduate student

Tobias Jonson

Currently Northeastern undergraduate student

Luke DeMarco

Currently Northeastern undergraduate student

Dominica Petrasova

Currently Northeastern undergraduate student

Oliver Stearns

Currently Northeastern undergraduate student

Isabelle Chambers

Currently Northeastern undergraduate student

Dylan Ingham

Currently Northeastern undergraduate student

SERVICE: SCIENTIFIC LEADERSHIP

- Organizing the *7th* [Single-Cell Proteomics Conference \(SCP2024\)](#) **May 28–29 2024**
- Organized the *6th* [Single-Cell Proteomics Conference \(SCP2023\)](#) **June 1–2 2023**
- Member of an advisory board setting priorities for CZI funding **Sept 8–9 2020**
- Organized the *5th* [Single-Cell Proteomics Conference \(SCP2022\)](#) **June 15–17 2022**
- Organized the *4th* [Single-Cell Proteomics Conference \(SCP2021\)](#) **Aug 16–18 2021**
- Member of a think tank on Functional Proteomics NIH (OD) **Sept 8–9 2020**
- Organized the third [Single-Cell Proteomics Conference \(SCP2020\)](#) **Aug 18–19 2020**
- Co-organized [Learning Meaningful Representations of Life](#) workshop at NeurIPS **Dec 13 2019**
- Member of a think-tank on single-cell analysis NIH (NHLBI) **Oct 17–18 2019**
- Organized the second [Single-Cell Proteomics Conference \(SCP2019\)](#) **June 10–12 2019**
- Chair for the fifth [Annual Single Cell Analysis USA Congress](#) **May 14 2019**
- Chair for the [Single Cell Proteomics session](#), HUPO **Oct 1–3 2018**
- Organized the first [Single-Cell Proteomics Conference \(SCP2018\)](#) **June 9–10 2018**

SERVICE:
PROFESSIONAL
ACTIVITIES

- **Peer-review:** Nature, Science, PNAS, Nature Molecular and Structural Biology, Molecular Biology of the Cell, PLoS, Nucleic Acids Research, Bioinformatics, Annals of Applied Statistics, Cell Reports, eLife, Nature Communications || [Publons Profile](#)
- **Editorial roles:** Editor for Journal of Proteome Research, eLife and PeerJ
- **Editorial advisory roles:** bioRxiv Affiliate, Member of the editorial board of Journal of Proteome Research and Analytical Chemistry

SERVICE:
DEPARTMENT
COMMITTEES

- Member of the tenure and promotion Committee **2022–2022**
- Head of BioE Search Committee (Recruited Miten Jain) **2021–2022**
- Member of the BioE Awards Committee **2020–2021**
- Member of the Search Committee for Barnett Institute Director **2018–2021**
- Member of the Departmental Faculty Search Committee **2016–2020**
- Member of the Graduate Student Admissions Committee **2017–2020**